# AccessRefinery: Fast Mining Concise Access Control Intents on Public Cloud

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Java 17

tests passing

```
<svg xmlns="http://www.w3.org/2000/svg" width="120" height="20" role="img" aria-label="li
```

paper FSE2026

## About AccessRefinery

**AccessRefinery** automatically mines access control intents from IAM (Identity and Access Management) policies. These intents help users verify policy correctness. Compared with AWS Access Analyzer, AccessRefinery accelerates mining by ~10-100x and reduces the number of intents by up to ~10x.

To accelerate intent mining, **AccessRefinery** uses our Multi-Theory Constraint Preprocessor (**MCP**) to speed up multi-round SMT solving by preprocessing constraints into bit-vector constraints.
For intent reduction, **AccessRefinery** computes a compact set that covers all mined intents by solving a minimum set-cover problem.
Moreover, we design **MCP** as a module separate from **AccessRefinery**, allowing other researchers to reuse **MCP** flexibly.

For technical details, see our FSE 2026 paper: AccessRefinery: Fast Mining Concise Access Control Intents on Public Cloud.

# Structure

Since AWS Access Analyzer is not open source and provides only a Command-Line Interface (CLI), we also reimplemented Access Analyzer for evaluation. We distinguish the two versions as the **Reimplemented Access Analyzer** and the **CLI-based Access Analyzer**.

- `data/`:
  - `Correctness/`: Dataset for correctness experiments.
  - `Scalability_05Keys/`: Synthetic dataset for scalability experiments.
  - `Scalability_06Keys/`: Synthetic dataset for scalability experiments.
- `accessrefinery/`: Implementation of AccessRefinery.
  - `bdd/`: Implementation of the binary decision diagram backend used by MCP.
  - `mcp/`: Implementation of the Multi-Theory Constraint Preprocessor (MCP).
  - `refinery/`: Implementation of intent mining and reduction.
- `baselines/`:
  - `accessanalyzer-reimpl`: Reimplementation of Access Analyzer.
  - `accessanalyzer-cli`: Scripts for invoking AWS Access Analyzer via CLI.
- `pom.xml`: Maven root configuration.
- `tools/`: Scripts for running the experiments.
- `docs/`: Stores documents such as auto-generated JavaDoc.
  - `mcp-javadoc`
  - `accessrefinery-javadoc`
- `paper_figures/`: Scripts for plotting the figures in the paper.
- `archive_results/`: Archived experimental results.

# Setup

## Prerequisites

See Requirements and Installation for detailed setup instructions (Linux, Java, Z3, and jq).

## Build

From the project root directory, run:

```
mvn clean package
```

The build generates the following JAR packages in `target/`:

- `mcp-1.0.jar` for **MCP**, which can be reused in other projects for fast multi-round SMT solving.
- `accessrefinery-1.0.jar` for **AccessRefinery**.
- `accessanalyzer-1.0.jar` for the reimplemented **Access Analyzer**.

For development in VS Code IDEA, see details.

# Using Multi-Theory Constraint Preprocessor (MCP)

**MCP** is a data structure for fast multi-round SMT solving. It supports regular expressions, IP prefixes/bit-vectors, ranges, and sets.

## Reuse in Another Project

Install `target/mcp-1.0.jar` into your local Maven repository:

```
mvn install:install-file \
    -Dfile=target/mcp-1.0.jar \
    -DgroupId=org.ants \
    -DartifactId=accessrefinery \
    -Dversion=1.0 \
    -Dpackaging=jar \
    -DgeneratePom=true
```

Then add the dependency to your `pom.xml`:

```
<dependencies>
    <dependency>
        <groupId>org.ants</groupId>
        <artifactId>accessrefinery</artifactId>
        <version>1.0</version>
    </dependency>
</dependencies>
```

# Example

This example follows the example in the paper (line 414).
Suppose we have the following IAM policy and a target intent, `Intent_6` (`Resource`: `dept*/user1.txt`, `IpAddress`: `112.0.0.0/24`).

```
{
    "Statement": [
        {
            "Effect": "Allow",
            "Resource": ["dept*/user1.txt", "dept1/user*.txt"],
            "Condition": {
                "IpAddress": {
                    "aws:SourceIp": ["112.0.0.0/24", "113.0.0.0/24"]
                }
            }
        },
        {
            "Effect": "Deny",
            "NotResource": "dept1/user*.txt",
            "Condition": {
                "IpAddress": {
                    "aws:SourceIp": "112.0.0.0/24"
                }
            }
        },
        {
            "Effect": "Deny",
            "NotResource": "dept*/user1.txt",
            "Condition": {
                "IpAddress": {
                    "aws:SourceIp" : "113.0.0.0/24"
                }
            }
        }
    ]
}
```

To check the satisfiability of three formulas, $\neg I_6 \wedge P \neg \mathrm{I6} \wedge \mathrm{P}$, $I_6 \wedge \neg P \mathrm{I6} \wedge \neg \mathrm{P}$, and $I_6 \wedge P \mathrm{I6} \wedge \mathrm{P}$, we use the following code based on **MCP**.

The example is also included in MCPFactoryTest.java and runs automatically during `mvn package`.

```java
package com.example;
import org.batfish.datamodel.Prefix;
import org.junit.Assert;
import org.mcp.core.MCPBitVector;
import org.mcp.core.MCPFactory;
import org.mcp.core.MCPFactory.MCPType;
import org.mcp.variables.statics.LabelType;

public class Main {
    public static void main(String[] args) {
        MCPFactory mcp = new MCPFactory(MCPType.BDD);
        mcp.addVar("Res", LabelType.REGEXP, "dept*/user1.txt");
        mcp.addVar("Res", LabelType.REGEXP, "dept1/user*.txt");
        mcp.addVar("IP", LabelType.PREFIX, Prefix.parse("112.0.0.0/24"));
        mcp.addVar("IP", LabelType.PREFIX, Prefix.parse("113.0.0.0/24"));
        mcp.updates();

        MCPBitVector res1 = mcp.getVar("Res", "dept*/user1.txt");
        MCPBitVector res2 = mcp.getVar("Res", "dept1/user*.txt");
        MCPBitVector ip1 = mcp.getVar("IP", Prefix.parse("112.0.0.0/24"));
        MCPBitVector ip2 = mcp.getVar("IP", Prefix.parse("113.0.0.0/24"));
        MCPBitVector s1 = (res1.or(res2)).and(ip1.or(ip2));
        MCPBitVector s2 = res1.not().and(ip1);
        MCPBitVector s3 = res2.not().and(ip2);
        MCPBitVector policy = s1.diff(s2).diff(s3);
        MCPBitVector intent6 = res1.and(ip1);

        // ¬I6∧P is satisfiable.
        Assert.assertTrue(!policy.and(intent6.not()).isZero());
        // I6∧¬P is unsatisfiable.
        Assert.assertTrue(policy.not().and(intent6).isZero());
        // I6∧P is satisfiable.
        Assert.assertTrue(!policy.and(intent6).isZero());
    }
}
```

# Using AccessRefinery

**AccessRefinery** builds on **MCP** for IAM intent mining and reduction. In this repository, **MCP** is already integrated into **AccessRefinery**, so you can use it directly without a separate installation.

To run **AccessRefinery**, use:

```
java -jar target/accessrefinery-1.0.jar [options]
```

**Command-line options:**

- `-h, --help` : Show help information.
- `-m, --mine` : Enable intent mining.
- `-r, --reduce` : Enable intent reduction.
- `-f, --file <path>` : Input path for policy files (must be under `data/`).
- `-s, --sat` : Use SAT to encode bit-vectors (default is BDD).
- `--round <number>` : Number of mining rounds (to reduce experimental bias).

**Example:**

```
java -jar target/accessrefinery-1.0.jar -m -r --round 1 -f data/Correctness
```

The command produces logs similar to the following:

```
[INFO] 2026-04-05 22:51:33 : ----------[ AccessRefinery Mode ]-------------
[INFO] 2026-04-05 22:51:33 : input  path: data/Correctness
[INFO] 2026-04-05 22:51:33 : output path: result/Correctness
[INFO] 2026-04-05 22:51:33 : ----------< 1th policy - 11_allow_allow_equal.json >---------
[INFO] 2026-04-05 22:51:33 : [1/6]  finish parser policy
[INFO] 2026-04-05 22:51:33 : [2/6]  finish ECs calculation
...
```

Results are generated in the `results/Correctness/` directory and include:

- `xxx.json`: The intents for each policy.

- `xxx.csv` : Statistics for multi-round SMT solving for each policy.
- `summary.txt` : Summary statistics for all policies in a folder.

In addition, one file is generated in the current path:

- `accessrefinery.log` : Records the running log.

# Evaluation Reproduction

This section describes how to reproduce the paper's conclusions. All archived results are available in `archive_results/`.

> We omit the results for the real-world datasets because of commercial restrictions.

We recommend skipping reproduction of results from the **reimplemented Access Analyzer** (because it takes a very long time) and the **CLI-based Access Analyzer** (because the setup is complex and requires AWS account registration, billing setup, and CLI credential configuration). We still provide instructions for developers.

- See details for the **Reimplemented Access Analyzer**.
- See details for the **CLI-based Access Analyzer**.

## Reproducing AccessRefinery Archived Results

The following scripts invoke `target/accessrefinery-1.0.jar` and reproduce the **AccessRefinery** results.

```
# The execution takes about 5min
sh tools/accessrefinery/running_bdd_miner.sh

# The execution takes about 5min
sh tools/accessrefinery/running_sat_miner.sh

# The execution takes about 10min
sh tools/accessrefinery/running_bdd_reducer.sh

# The execution takes about 1h
sh tools/accessrefinery/running_sat_reducer.sh

# The execution takes about 10 seconds.
sh tools/accessrefinery/running_batch_compare.sh
```

The following directories are generated under `results/`:

- `accessrefinery_bdd_miner_10rs/`: Intent mining results for 10 rounds with JavaBDD.
- `accessrefinery_sat_miner_10rs/`: Intent mining results for 10 rounds with MiniSAT.
- `accessrefinery_bdd_reducer_10rs/`: Intent mining and reduction results for 10 rounds with JavaBDD.
- `accessrefinery_sat_reducer_3rs/`: Intent mining and reduction results for 3 rounds with MiniSAT.
- `compare_accessrefinery_with_accessanalyzer_cli/`: Comparison logs between **AccessRefinery** and the **CLI-based Access Analyzer**, and between **AccessRefinery (BDD)** and **AccessRefinery (SAT)**.

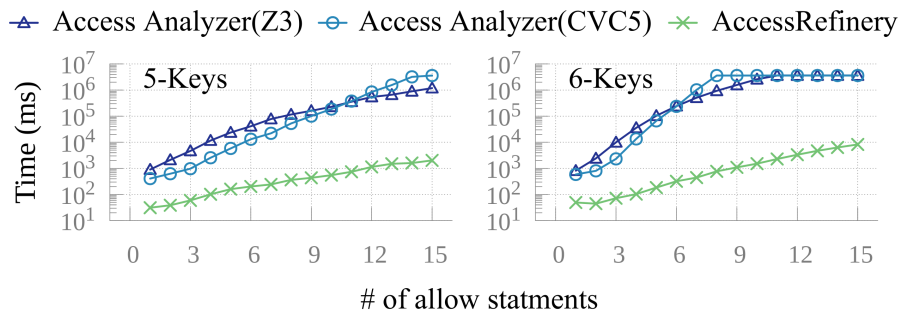
> Because the MiniSAT-based reduction experiment runs very slowly, we report only three rounds for that experiment.

# Mapping results to paper conclusions

This section presents two representative conclusions. See details for the full explanation.

1. **AccessRefinery** is 10-100x faster than **Access Analyzer** for intent

mining.




**Required logs**:

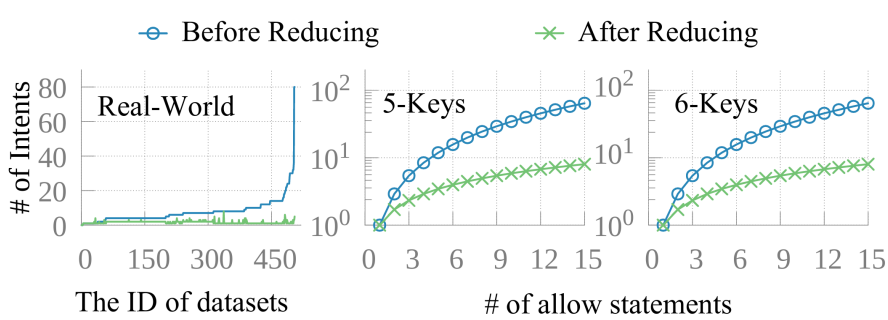
- `accessrefinery_bdd_miner_10rs/`
  - `Scalability_05Keys/summary.txt`
  - `Scalability_06Keys/summary.txt`

`TotalTimeAverage` denotes the average runtime over 10 rounds for **AccessRefinery**.

- `accessanalyzer_z3_unreduced/`
  - `Scalability_05Keys/summary.csv`
  - `Scalability_06Keys/summary.csv`
- `accessanalyzer_cvc5_unreduced/`
  - `Scalability_05Keys/summary.csv`
  - `Scalability_06Keys/summary.csv`

`Total Time (s)` denotes the runtime for the **reimplemented Access Analyzer**. We report one round because it is slow.

2. **AccessRefinery** can reduce the number of intents by up to 10x after reduction.




**Required logs**:

- `accessrefinery_bdd_reducer_10rs/`
    - `Scalability_05Keys/summary.txt`
    - `Scalability_06Keys/summary.txt`

`NumberMCI` denotes the number of intents before reduction, and `NumberRRI` denotes the number of intents after reduction.

## Plotting the Figures

We also provide scripts for drawing the figures in the paper. The following commands install `gnuplot` and generate all figures used in the experiments.

```
sudo apt install gnuplot
cd paper_figures
sh draw.sh
```

The generated figures are saved in `paper_figures/results/`.

# Contact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